

## SAFETY DATA SHEET

Revision Date 01-Apr-2024

Revision Number 3

### 1. Identification

**Product Name** Graphite conductive adhesive, alcohol based

**Cat No. :** 41213

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                                   |            |
|-------------------------------------------------------------------|------------|
| Flammable liquids                                                 | Category 2 |
| Serious Eye Damage/Eye Irritation                                 | Category 1 |
| Specific target organ toxicity (single exposure)                  | Category 3 |
| Target Organs - Respiratory system, Central nervous system (CNS). |            |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Highly flammable liquid and vapor  
Causes serious eye damage  
May cause respiratory irritation  
May cause drowsiness or dizziness

**Precautionary Statements****Prevention**

Wear protective gloves/protective clothing/eye protection/face protection

Avoid breathing dust/fume/gas/mist/vapors/spray

Use only outdoors or in a well-ventilated area

Keep away from heat/sparks/open flames/hot surfaces. - No smoking

Keep container tightly closed

Ground/bond container and receiving equipment

Use explosion-proof electrical/ventilating/lighting equipment

Use only non-sparking tools

Take precautionary measures against static discharge

Keep cool

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

Call a POISON CENTER or doctor/physician if you feel unwell

**Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER or doctor/physician

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store in a well-ventilated place. Keep container tightly closed

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component                         | CAS No    | Weight % |
|-----------------------------------|-----------|----------|
| Isopropyl alcohol                 | 67-63-0   | 50       |
| Propylene glycol monomethyl ether | 107-98-2  | 20       |
| Graphite                          | 7782-42-5 | 20       |
| n-Butyl alcohol                   | 71-36-3   | 5        |
| Proprietary thickener             | N/A       | 5        |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

|                                            |                                                                                                                                                                                          |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                             |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                                                             |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Causes eye burns. Causes severe eye damage. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                    |

## 5. Fire-fighting measures

|                                         |                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Powder. Water spray. In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                                                                               |
| <b>Flash Point</b>                      | 11 °C / 51.8 °F                                                                                                                                                                                                        |
| <b>Method -</b>                         | No information available                                                                                                                                                                                               |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                                                                                               |
| <b>Explosion Limits</b>                 |                                                                                                                                                                                                                        |
| <b>Upper</b>                            | No data available                                                                                                                                                                                                      |
| <b>Lower</b>                            | No data available                                                                                                                                                                                                      |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                                                               |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                                                               |

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 2             | 3                   | 0                  | -                       |

## 6. Accidental release measures

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                            |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Take precautionary measures against static discharges.

**Storage.**

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame. Incompatible Materials. Oxidizing agent.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component                         | ACGIH TLV                     | OSHA PEL                                                                                                                                                                          | NIOSH                                                                                                         | Mexico OEL (TWA)                                      |
|-----------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Isopropyl alcohol                 | TWA: 200 ppm<br>STEL: 400 ppm | (Vacated) TWA: 400 ppm<br>(Vacated) TWA: 980 mg/m <sup>3</sup><br>(Vacated) STEL: 500 ppm<br>(Vacated) STEL: 1225 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup> | IDLH: 2000 ppm<br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup><br>STEL: 500 ppm<br>STEL: 1225 mg/m <sup>3</sup> | TWA: 200 ppm<br>STEL: 400 ppm                         |
| Propylene glycol monomethyl ether | TWA: 50 ppm<br>STEL: 100 ppm  | (Vacated) TWA: 100 ppm<br>(Vacated) TWA: 360 mg/m <sup>3</sup><br>(Vacated) STEL: 150 ppm<br>(Vacated) STEL: 540 mg/m <sup>3</sup>                                                | TWA: 100 ppm<br>TWA: 360 mg/m <sup>3</sup><br>STEL: 150 ppm<br>STEL: 540 mg/m <sup>3</sup>                    | TWA: 100 ppm<br>STEL: 150 ppm                         |
| Graphite                          | TWA: 2 mg/m <sup>3</sup>      | (Vacated) TWA: 2.5 mg/m <sup>3</sup><br>(Vacated) TWA: 10 mg/m <sup>3</sup><br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>TWA: 15 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup>        | IDLH: 1250 mg/m <sup>3</sup><br>TWA: 2.5 mg/m <sup>3</sup>                                                    | TWA: 2 mg/m <sup>3</sup><br>TWA: 10 mg/m <sup>3</sup> |
| n-Butyl alcohol                   | TWA: 20 ppm                   | Skin<br>(Vacated) Ceiling: 50 ppm<br>(Vacated) Ceiling: 150 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 300 mg/m <sup>3</sup>                                                       | IDLH: 1400 ppm<br>Ceiling: 50 ppm<br>Ceiling: 150 mg/m <sup>3</sup>                                           | TWA: 20 ppm                                           |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:**

Organic gases and vapours filter. Type A. Brown. conforming to EN14387.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

**Physical State**

Liquid Viscous liquid

|                                        |                          |
|----------------------------------------|--------------------------|
| Appearance                             | Black                    |
| Odor                                   | No information available |
| Odor Threshold                         | No information available |
| pH                                     | No information available |
| Melting Point/Range                    | No data available        |
| Boiling Point/Range                    | 82 °C / 179.6 °F         |
| Flash Point                            | 11 °C / 51.8 °F          |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | 23 hPa @ 20 °C           |
| Vapor Density                          | No information available |
| Specific Gravity                       | 0.9 g/cm3                |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

|                                  |                                                                   |
|----------------------------------|-------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                        |
| Stability                        | Stable under normal conditions.                                   |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Oxidizing agent                                                   |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )           |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                          |
| Hazardous Reactions              | None under normal processing.                                     |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

**Oral LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Dermal LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Vapor LC50** Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

#### Component Information

| Component                         | LD50 Oral                                  | LD50 Dermal                  | LC50 Inhalation                           |
|-----------------------------------|--------------------------------------------|------------------------------|-------------------------------------------|
| Isopropyl alcohol                 | 5045 mg/kg ( Rat )<br>3600 mg/kg ( Mouse ) | 12800 mg/kg ( Rat )          | 72.6 mg/L ( Rat ) 4 h                     |
| Propylene glycol monomethyl ether | LD50 = 5000 mg/kg ( Rat )                  | LD50 = 13 g/kg ( Rabbit )    | LC50 > 7559 ppm ( Rat ) 6 h               |
| Graphite                          | Not listed                                 | Not listed                   | LC50 > 2000 mg/m <sup>3</sup> ( Rat ) 4 h |
| n-Butyl alcohol                   | LD50 = 700 mg/kg ( Rat )                   | LD50 = 3402 mg/kg ( Rabbit ) | LC50 > 8000 ppm ( Rat ) 4 h               |

**Toxicologically Synergistic** No information available

#### Products

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                         | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------------------|-----------|------------|------------|------------|------------|------------|
| Isopropyl alcohol                 | 67-63-0   | Not listed | Not listed | Not listed | Not listed | Not listed |
| Propylene glycol monomethyl ether | 107-98-2  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Graphite                          | 7782-42-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| n-Butyl alcohol                   | 71-36-3   | Not listed | Not listed | Not listed | Not listed | Not listed |
| Proprietary thickener             | N/A       | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system Central nervous system (CNS)

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

| Component                         | Freshwater Algae                                                                                                                                                               | Freshwater Fish                                                                                                                                                                                                                         | Microtox                                                                                              | Water Flea                                                                                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Isopropyl alcohol                 | EC50: > 1000 mg/L, 72h (Desmodesmus subspicatus)<br>EC50: > 1000 mg/L, 96h (Desmodesmus subspicatus)                                                                           | LC50: = 9640 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: > 1400000 µg/L, 96h (Lepomis macrochirus)<br>LC50: = 11130 mg/L, 96h static (Pimephales promelas)<br>LC50: = 10000000 µg/L, 96h (Daphnia)                            | = 35390 mg/L EC50 Photobacterium phosphoreum 5 min                                                    | 13299 mg/L EC50 = 48 h<br>9714 mg/L EC50 = 24 h                                                                                                         |
| Propylene glycol monomethyl ether | Not listed                                                                                                                                                                     | LC50: = 20.8 g/L, 96h static (Pimephales promelas)                                                                                                                                                                                      | Not listed                                                                                            | EC50: = 23300 mg/L, 48h (Daphnia magna)                                                                                                                 |
| Graphite                          | Not listed                                                                                                                                                                     | LC50: > 100 mg/L, 96h semi-static (Danio rerio)                                                                                                                                                                                         | Not listed                                                                                            | Not listed                                                                                                                                              |
| n-Butyl alcohol                   | EC50: 225 mg/L, 96h (Pseudokirchneriella subcapitata) OECD Guideline 201<br>EC50: > 500 mg/L, 72h (Desmodesmus subspicatus)<br>EC50: > 500 mg/L, 96h (Desmodesmus subspicatus) | LC50: 1376 mg/L, 96h (Pimephales promelas) OECD Guideline 203 :<br>100000 - 500000 µg/L, 96h static (Lepomis macrochirus)<br>LC50: = 1740 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 1910000 µg/L, 96h static (Pimephales) | EC50 = 2041.4 mg/L 5 min<br>EC50 = 2186 mg/L 30 min<br>EC50 = 3980 mg/L 24 h<br>EC50 = 4400 mg/L 17 h | EC50: 1328 mg/L, 48h (Daphnia magna) OECD Guideline 202<br>EC50: 1897 - 2072 mg/L, 48h Static (Daphnia magna)<br>EC50: = 1983 mg/L, 48h (Daphnia magna) |

|  |  |                                                                             |  |  |
|--|--|-----------------------------------------------------------------------------|--|--|
|  |  | promelas)<br>LC50: 1730 - 1910 mg/L,<br>96h static (Pimephales<br>promelas) |  |  |
|--|--|-----------------------------------------------------------------------------|--|--|

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its volatility.

| Component                         | log Pow |
|-----------------------------------|---------|
| Isopropyl alcohol                 | 0.05    |
| Propylene glycol monomethyl ether | 1       |
| n-Butyl alcohol                   | 1       |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component                 | RCRA - U Series Wastes | RCRA - P Series Wastes |
|---------------------------|------------------------|------------------------|
| n-Butyl alcohol - 71-36-3 | U031                   | -                      |

### 14. Transport information

#### DOT

UN-No UN1133  
 Proper Shipping Name Adhesives  
 Hazard Class 3  
 Packing Group II

#### TDG

UN-No UN1133  
 Proper Shipping Name ADHESIVES  
 Hazard Class 3  
 Packing Group II

#### IATA

UN-No UN1133  
 Proper Shipping Name ADHESIVES  
 Hazard Class 3  
 Packing Group II

#### IMDG/IMO

UN-No UN1133  
 Proper Shipping Name ADHESIVES  
 Hazard Class 3  
 Packing Group II

### 15. Regulatory information

#### United States of America Inventory

| Component                         | CAS No    | TSCA | TSCA Inventory notification -<br>Active-Inactive | TSCA - EPA Regulatory<br>Flags |
|-----------------------------------|-----------|------|--------------------------------------------------|--------------------------------|
| Isopropyl alcohol                 | 67-63-0   | X    | ACTIVE                                           | -                              |
| Propylene glycol monomethyl ether | 107-98-2  | X    | ACTIVE                                           | -                              |
| Graphite                          | 7782-42-5 | X    | ACTIVE                                           | -                              |
| n-Butyl alcohol                   | 71-36-3   | X    | ACTIVE                                           | -                              |
| Proprietary thickener             | N/A       | -    | -                                                | -                              |

#### **Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed  
 '-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b) - Notices of Export**

Not applicable

**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                         | CAS No    | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|-----------------------------------|-----------|-----|------|-----------|-------|------|------|------|-------|----------|
| Isopropyl alcohol                 | 67-63-0   | X   | -    | 200-661-7 | X     | X    | X    | X    | X     | KE-29363 |
| Propylene glycol monomethyl ether | 107-98-2  | X   | -    | 203-539-1 | X     | X    | X    | X    | X     | KE-23379 |
| Graphite                          | 7782-42-5 | X   | -    | 231-955-3 | X     | -    |      | X    | X     | KE-18101 |
| n-Butyl alcohol                   | 71-36-3   | X   | -    | 200-751-6 | X     | X    | X    | X    | X     | KE-03867 |
| Proprietary thickener             | N/A       | -   | -    | -         | -     | -    |      | -    | -     | -        |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations**
**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component         | CAS No  | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|-------------------|---------|----------|-------------------------------|---------------------------------|
| Isopropyl alcohol | 67-63-0 | 50       | 1.0 %                         | -                               |
| n-Butyl alcohol   | 71-36-3 | 5        | 1.0 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

Not applicable

**Clean Air Act**

Not applicable

**OSHA - Occupational Safety and Health Administration**

Not applicable

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component       | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|-----------------|--------------------------|-------------------------------------------|-------------------------------|
| n-Butyl alcohol | 5000 lb                  | -                                         | 5000 lb<br>2270 kg            |

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component                         | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-----------------------------------|---------------|------------|--------------|----------|--------------|
| Isopropyl alcohol                 | X             | X          | X            | -        | X            |
| Propylene glycol monomethyl ether | X             | X          | X            | X        | X            |
| Graphite                          | X             | X          | X            | -        | -            |
| n-Butyl alcohol                   | X             | X          | X            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y  
 DOT Marine Pollutant N  
 DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations****Mexico - Grade**

No information available

**Authorisation/Restrictions according to EU REACH**

| Component                         | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Isopropyl alcohol                 | 67-63-0   | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Propylene glycol monomethyl ether | 107-98-2  | -                                                                   | -                                                                             | -                                                                                                     |
| Graphite                          | 7782-42-5 | -                                                                   | -                                                                             | -                                                                                                     |
| n-Butyl alcohol                   | 71-36-3   | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Proprietary thickener             | N/A       | -                                                                   | -                                                                             | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                         | CAS No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Isopropyl alcohol                 | 67-63-0   | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Propylene glycol monomethyl ether | 107-98-2  | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Graphite                          | 7782-42-5 | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| n-Butyl alcohol                   | 71-36-3   | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Proprietary thickener             | N/A       | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component                         | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Isopropyl alcohol                 | 67-63-0   | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y42                      |
| Propylene glycol monomethyl ether | 107-98-2  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Graphite                          | 7782-42-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| n-Butyl alcohol                   | 71-36-3   | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Proprietary thickener             | N/A       | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Revision Date**

01-Apr-2024

**Print Date**

01-Apr-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**